For every i, let

$$\psi_i = \frac{\dim V_i}{|G|} \sum_{g \in G} \chi_{V_i}(g) \cdot g^{-1} \in \mathbb{C}[G].$$

(i) Prove that ψ_i acts on V_j as the identity if $j = i$, and as the null map if $j \neq i$.

(ii) Prove that ψ_i are **idempotents**; i.e., $\psi_i^2 = \psi_i$ for any i, and $\psi_i \psi_j = 0$ for any $i \neq j$.

Hint: In (i), notice that ψ_i commutes with any element of $k[G]$ and thus acts on V_j as an intertwining operator. Corollary 2.3.10 thus yields that ψ_i acts on V_j as a scalar. Compute this scalar by taking its trace in V_j.

Remark 4.5.3. We see that characters of irreducible complex representations of G can be defined without mentioning irreducible representations. Namely, equip the space $F(G, \mathbb{C})$ of complex-valued functions on G with the convolution product

$$(f * g)(z) = \sum_{x,y \in G : xy = z} f(x) g(y).$$

This product turns $F(G, \mathbb{C})$ into an associative algebra, with unit δ_e (the characteristic function of the unit $e \in G$), and the space of class functions $F_c(G, \mathbb{C})$ is a commutative subalgebra. Then one can define renormalized characters $\tilde{\chi}_i \in F_c(G, \mathbb{C})$ to be the primitive idempotents in this algebra, i.e., solutions of the equation $f * f = f$ which cannot be decomposed into a sum of other nonzero solutions. Then one can define the characters by the formula

$$\chi_i(g) = \sqrt{\frac{|G|}{\tilde{\chi}_i(1)}} \tilde{\chi}_i(g)$$

(check it!). This is, essentially, how Frobenius defined characters (see [**Cu**], equation (7)). Note that Frobenius defined representations at approximately the same time, but for some time it was not clear that there is a simple relation between irreducible representations and characters (namely, that irreducible characters are simply traces of group elements in irreducible representations). Even today, many group theorists sometimes talk of irreducible characters of a finite group rather than irreducible representations.